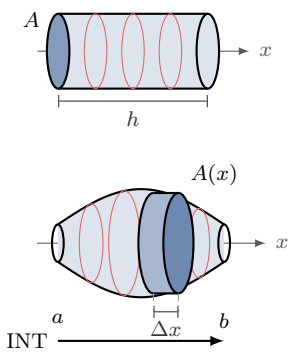
L4.2a Volumes by Slicing

Disks and cross-sections — what you need to know

VOLUME BY SLICING



Cylinder. Every slice perpendicular to the x-axis is the same disk, area $A = \pi r^2$. So the volume is the base area times the height:

$$V = Ah = \pi r^2 h$$

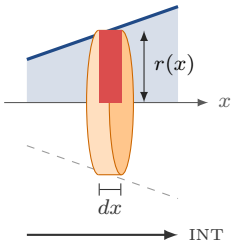
Generalize. When the slices change, cut the solid by planes perpendicular to the x-axis. The slice at x is a little chunk of area $A(x)$ and thickness Δx, so $V(S_i) \approx A(x_i^*)\,\Delta x$. Add the chunks:

$$\boxed{V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx = \int dV}$$

$dV = A(x)\,dx$ or $A(y)\,dy$ is the volume of one thin chunk.

"the definition of volume: add up the little chunks dV, each one area times thickness"

VOLUME BY DISKS



When a region is spun about an axis, every chunk dV is a thin disk.

① **Draw a representative disk:** a thin rectangle **perpendicular** to the spin axis, from the axis to the curve, and the disk dV it sweeps. Label its radius $r(x)$ or $r(y)$: the length of the rectangle.

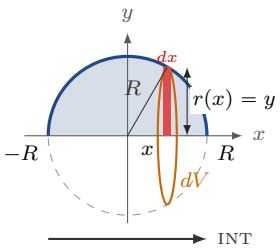
② **Path of integration and limits:** along the spin axis — dx for a horizontal axis, dy for a vertical one — from where the rectangles start to where they stop.

③ **A formula for the radius,** $r(x)$ or $r(y)$, in the variable of the path.

④ **The chunk:** $dV = A(x)\,dx = \pi\, r(x)^2\,dx$, or $A(y)\,dy = \pi\, r(y)^2\,dy$.

$$\boxed{V = \int_{\text{path}} dV}$$

VOLUME OF A SPHERE



Ex. Show that $V_{\text{sphere}} = \frac{4}{3}\pi R^3$.

① **Draw a representative disk:** an upright rectangle from the x-axis up to the circle $x^2 + y^2 = R^2$, the disk dV, and its radius $r(x)$.

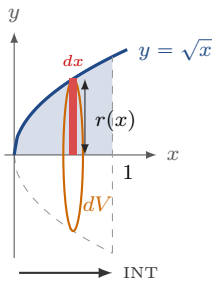
② **Path and limits:** dx, from $-R$ to R.

③ **Radius:** from the triangle, $R^2 = r(x)^2 + x^2$, so $r(x)^2 = R^2 - x^2$.

④ **Chunk:** $dV = A(x)\,dx = \pi\, r(x)^2\,dx = \pi\big(R^2 - x^2\big)\,dx$.

$$V = \int_{\text{path}} dV = \int_{-R}^{R} \pi\big(R^2 - x^2\big)\,dx = 2\pi \int_0^R R^2 - x^2\,dx = 2\pi\Big[R^3 - \tfrac{R^3}{3}\Big] = \boxed{\tfrac{4}{3}\pi R^3}$$

DISKS ABOUT THE x-AXIS



Ex. Rotate the region under $y = \sqrt{x}$, $0 \leqslant x \leqslant 1$, about the x-axis.

① **Draw a representative disk:** an upright rectangle from the axis up to $y = \sqrt{x}$, the disk dV, and its radius $r(x)$.

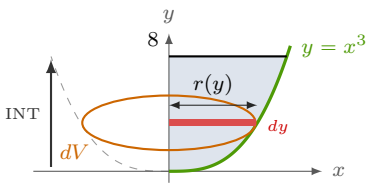
② **Path and limits:** dx, from 0 to 1.

③ **Radius:** $r(x)$ is y, so $r(x) = \sqrt{x}$.

④ **Chunk:** $dV = A(x)\,dx = \pi\left(\sqrt{x}\right)^2 dx = \pi x\,dx$.

$$V = \int_{\text{path}} dV = \int_0^1 \pi x\,dx = \pi\left(\frac{x^2}{2}\right)\Big|_0^1 = \boxed{\frac{\pi}{2}}$$

DISKS ABOUT THE y-AXIS



Ex. Rotate the region bounded by $y = x^3$, $y = 8$ and $x = 0$ about the y-axis.

① **Draw a representative disk:** a sideways rectangle from the y-axis across to the curve, the disk dV, and its radius $r(y)$.

② **Path and limits:** dy, from 0 to 8.

③ **Radius:** $r(y) = x$, but it is needed in terms of y. Invertible on the domain? Yes: $r(y) = y^{1/3}$.

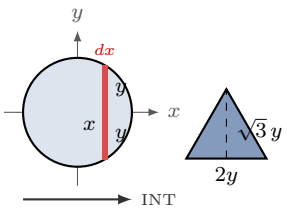
④ **Chunk:** $dV = A(y)\,dy = \pi\,r(y)^2\,dy = \pi y^{2/3}\,dy$.

$$V = \int_{\text{path}} dV = \pi\int_0^8 y^{2/3}\,dy = \frac{3\pi}{5}\,y^{5/3}\Big|_0^8 = \frac{3\pi}{5}\cdot 2^5 = \boxed{\frac{96\pi}{5}}$$

KNOWN CROSS-SECTIONS

Given the base and the shape of every slice, there is no radius: ① draws the slice and builds the whole $A(x)$ from the shape in one step, ② is the path and limits, ③ integrates.

square: $A = s^2$ **semicircle,** diameter s: $A = \frac{\pi}{8}s^2$ **equilateral triangle:** $A = \frac{1}{2}s\cdot\frac{\sqrt{3}}{2}s = \frac{\sqrt{3}}{4}s^2$



Ex. The base is the disk $x^2 + y^2 \leqslant 1$. Cross-sections perpendicular to the x-axis are equilateral triangles.

① **Draw a representative slice** $\perp$ the x-axis, the triangle dV standing on it, and $A(x)$: the side is $2y$, the height $\sqrt{3}\,y$, so $A = \sqrt{3}\,y^2$. Switch to the x variable: $y^2 = 1 - x^2$, $A(x) = \sqrt{3}\left(1 - x^2\right)$.

② **Path and limits:** dx, from -1 to 1.

③ $V = \int_{\text{path}} dV = \int_{-1}^1 \sqrt{3}\left(1 - x^2\right)dx = 2\sqrt{3}\left(1 - \frac{1}{3}\right) = \boxed{\frac{4\sqrt{3}}{3}}$